

REVIEW AND RATING TABLE MANAGEMENT

Project Title: Myntra – E-Commerce database Management System Skill

1.Database design

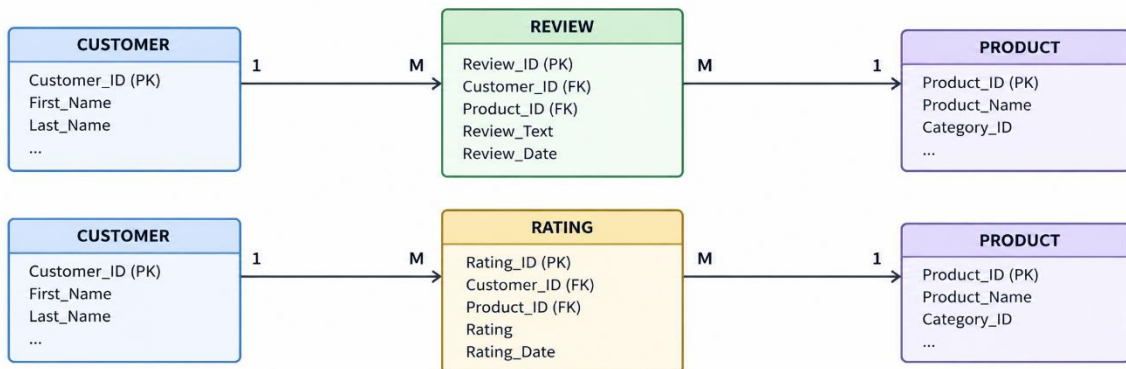
1.1Review Table

Attribute	Data Type	Key	Description
Review_ID	INT	PK	Unique review ID
Customer_ID	INT	FK	Identifies the customer who gives the review
Product_ID	INT	FK	Identifies the product being reviewed
Review_Text	VARCHAR2(500)	-	Textual feedback given by the customer
Review_Date	DATE	-	Date on which the review was submitted

1.2 Rating Table

Attribute	Data Type	Key	Description
Rating_ID	INT	PK	Unique rating ID
Customer_ID	INT	FK	Identifies the customer who gives the rating
Product_ID	INT	FK	Identifies the rated product
Rating	INT	-	Numerical rating given by the customer from 1 to 5
Rating_Date	DATE	-	Date on which the rating was submitted

2. Table Relationships



3. Table Creation

3.1 Create Review Table

```
SQL> CREATE TABLE Review (
2     Review_ID INT PRIMARY KEY,
3     Customer_ID INT REFERENCES Customer(Customer_ID),
4     Product_ID INT REFERENCES Product(Product_ID),
5     Review_Text VARCHAR2(500) NOT NULL,
6     Review_Date DATE NOT NULL
7 );
```

Table created.

3.2 Create Rating Table

```
SQL> CREATE TABLE Rating (
2     Rating_ID INT PRIMARY KEY,
3     Customer_ID INT REFERENCES Customer(Customer_ID),
4     Product_ID INT REFERENCES Product(Product_ID),
5     Rating INT NOT NULL,
6     Rating_Date DATE NOT NULL
7 );
```

Table created.

4. Review Data Insertion

```
SQL> INSERT INTO Review
  2  VALUES (801, 101, 201,
  3           'Good quality cotton shirt and comfortable to wear.',
  4           TO_DATE('17-SEP-2026', 'DD-MON-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES (802, 102, 202,
  3           'Beautiful floral dress with good fitting.',
  4           TO_DATE('18-SEP-2026', 'DD-MON-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES (803, 103, 203,
  3           'Running shoes are comfortable and suitable for daily use.',
  4           TO_DATE('19-SEP-2026', 'DD-MON-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES (804, 104, 204,
  3           'The watch looks stylish and works perfectly.',
  4           TO_DATE('20-SEP-2026', 'DD-MON-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES (805, 105, 205,
  3           'The backpack has good space and strong quality.',
  4           TO_DATE('21-SEP-2026', 'DD-MON-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES (806, 106, 201,
  3           'The shirt material is good but the size is slightly loose.',
  4           TO_DATE('22-SEP-2026', 'DD-MON-YYYY'));

1 row created.
```

5. Rating Data Insertion

```
SQL> INSERT INTO Rating
  2 VALUES (901, 101, 201, 4,
  3          TO_DATE('17-SEP-2026', 'DD-MON-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Rating
  2 VALUES (902, 102, 202, 5,
  3          TO_DATE('18-SEP-2026', 'DD-MON-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Rating
  2 VALUES (903, 103, 203, 5,
  3          TO_DATE('19-SEP-2026', 'DD-MON-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Rating
  2 VALUES (904, 104, 204, 4,
  3          TO_DATE('20-SEP-2026', 'DD-MON-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Rating
  2 VALUES (905, 105, 205, 5,
  3          TO_DATE('21-SEP-2026', 'DD-MON-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Rating
  2 VALUES (906, 106, 201, 3,
  3          TO_DATE('22-SEP-2026', 'DD-MON-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Rating
  2 VALUES (907, 107, 203, 4,
  3          TO_DATE('23-SEP-2026', 'DD-MON-YYYY'));

1 row created.
```

6. Retrieve Product Review Details

```
SQL> SELECT
  2     r.Review_ID,
  3     c.Customer_ID,
  4     c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  5     p.Product_ID,
  6     p.Product_Name,
  7     r.Review_Text,
  8     r.Review_Date
  9 FROM Review r
 10 JOIN Customer c
 11     ON r.Customer_ID = c.Customer_ID
 12 JOIN Product p
 13     ON r.Product_ID = p.Product_ID
 14 ORDER BY r.Review_Date DESC;
```

REVIEW_ID CUSTOMER_ID

CUSTOMER_NAME

PRODUCT_ID

PRODUCT_NAME

REVIEW_TEXT

REVIEW_DA

808 108

REVIEW_ID CUSTOMER_ID

CUSTOMER_NAME

PRODUCT_ID

PRODUCT_NAME

REVIEW_TEXT

REVIEW_DA

Meena Krishnan

REVIEW_ID CUSTOMER_ID

CUSTOMER_NAME

PRODUCT_ID

7. Calculate Average Product Ratings

This uses the AVG() aggregate function.

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     AVG(r.Rating) AS Average_Rating
  5 FROM Product p
  6 JOIN Rating r
  7   ON p.Product_ID = r.Product_ID
  8 GROUP BY
  9     p.Product_ID,
 10     p.Product_Name
 11 ORDER BY Average_Rating DESC;
```

PRODUCT_ID

PRODUCT_NAME

AVERAGE_RATING

202
Floral Dress
5

203
Running Shoes
4.5

PRODUCT_ID

PRODUCT_NAME

AVERAGE_RATING

205
Backpack
4.5

204
Analog Watch

PRODUCT_ID

PRODUCT_NAME

AVERAGE_RATING
4

8. Product Rating Report

```

SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     AVG(r.Rating) AS Average_Rating
  5 FROM Product p
  6 JOIN Rating r
  7   ON p.Product_ID = r.Product_ID
  8 GROUP BY
  9     p.Product_ID,
 10     p.Product_Name
 11 ORDER BY Average_Rating DESC;

```

PRODUCT_ID

PRODUCT_NAME

AVERAGE_RATING

202
Floral Dress 5

203
Running Shoes 4.5

PRODUCT_ID

PRODUCT_NAME

AVERAGE_RATING

205
Backpack 4.5

204
Analog Watch

PRODUCT_ID

PRODUCT_NAME

AVERAGE_RATING

4
201

9. Product Rating Report

```
SQL> SELECT
2     r.Rating_ID,
3     p.Product_ID,
4     p.Product_Name,
5     r.Rating,
6     r.Rating_Date
7 FROM Rating r
8 JOIN Product p
9     ON r.Product_ID = p.Product_ID
10 ORDER BY r.Rating_Date DESC;
```

```
RATING_ID PRODUCT_ID
```

```
PRODUCT_NAME
```

```
RATING RATING_DA
```

```
908 205
```

```
Backpack
```

```
4 24-SEP-26
```

```
907 203
```

```
Running Shoes
```

```
4 23-SEP-26
```

```
RATING_ID PRODUCT_ID
```

```
PRODUCT_NAME
```

```
RATING RATING_DA
```

```
906 201
```

```
Cotton Shirt
```

```
3 22-SEP-26
```

```
905 205
```

```
Backpack
```

```
RATING_ID PRODUCT_ID
```

```
PRODUCT_NAME
```

```
RATING RATING_DA
```

```
5 21-SEP-26
```

```
904 204
```

```
Analog Watch
```